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COMPUTER VISION & AI

Vision Transformers

A PRESENTATION BY
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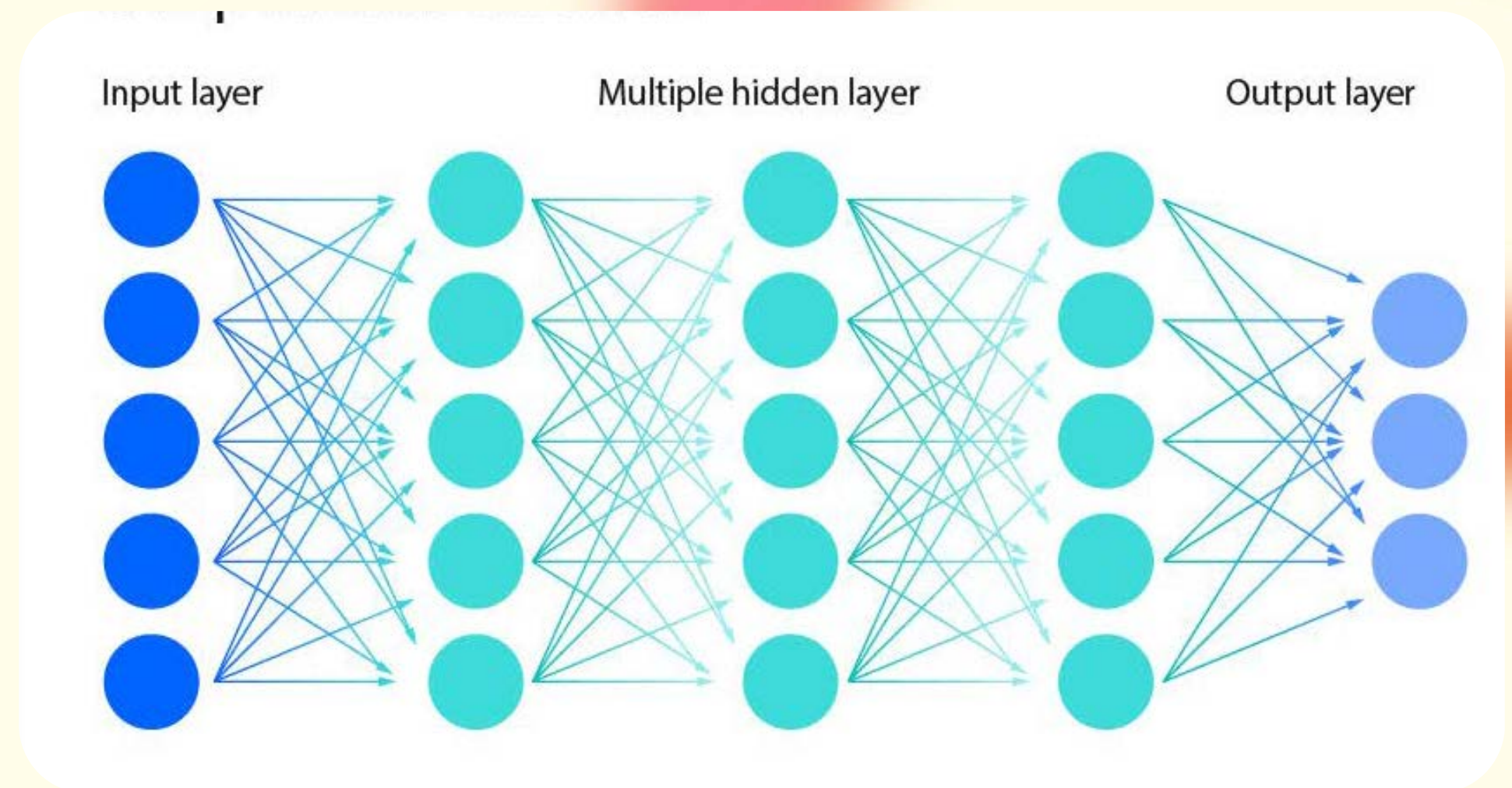
08 LIMITATIONS & COMPARISON

Foundation

NEURAL NETWORK

= interconnected computing nodes, or "neurons"

- Inspired by the brain's structure
- They learn patterns through adjusting **weights** (parameters) based on data
- CV applies neural networks to image data to extract **meaning**



SOTA before

CNN (Convolutional Neural Networks)

- Feed-forward architecture using filters and pooling
- For image recognition, classification, face detection and spatial data
- Use small learnable filters across images that detect spatial hierarchies of features^[1,3]

RNN (Recurrent Neural Networks)

- Recurrent, feeds results back into the network
- For speech recognition, time series forecasting and sequential data^[3]
- Incorporate feedback loops that allow information to persist across time steps

THE REVOLUTION OF

Transformers

Transformers

"Attention Is All You Need" Paper (2017 - Google) ^[4]



Modern neural network architecture

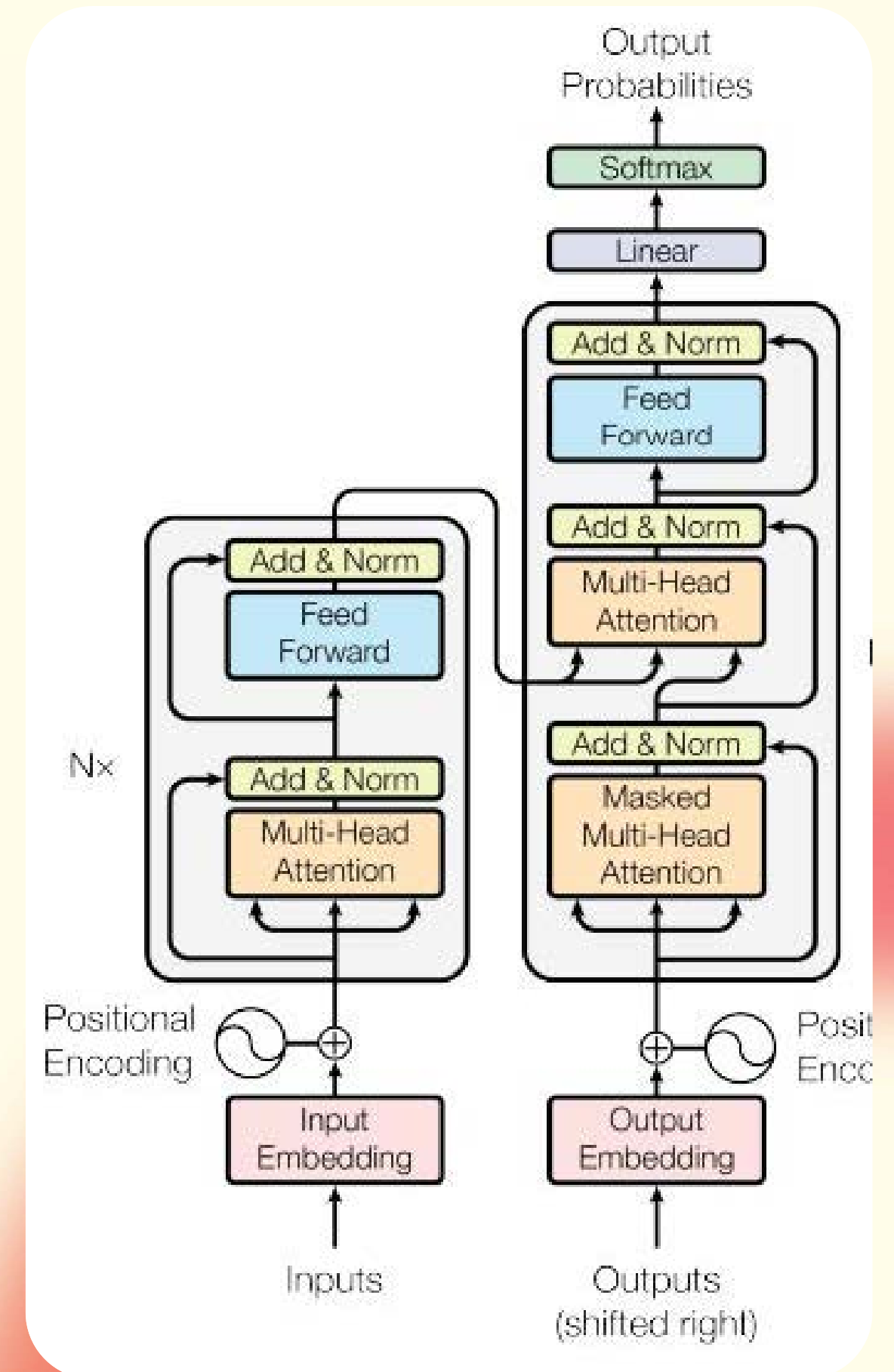
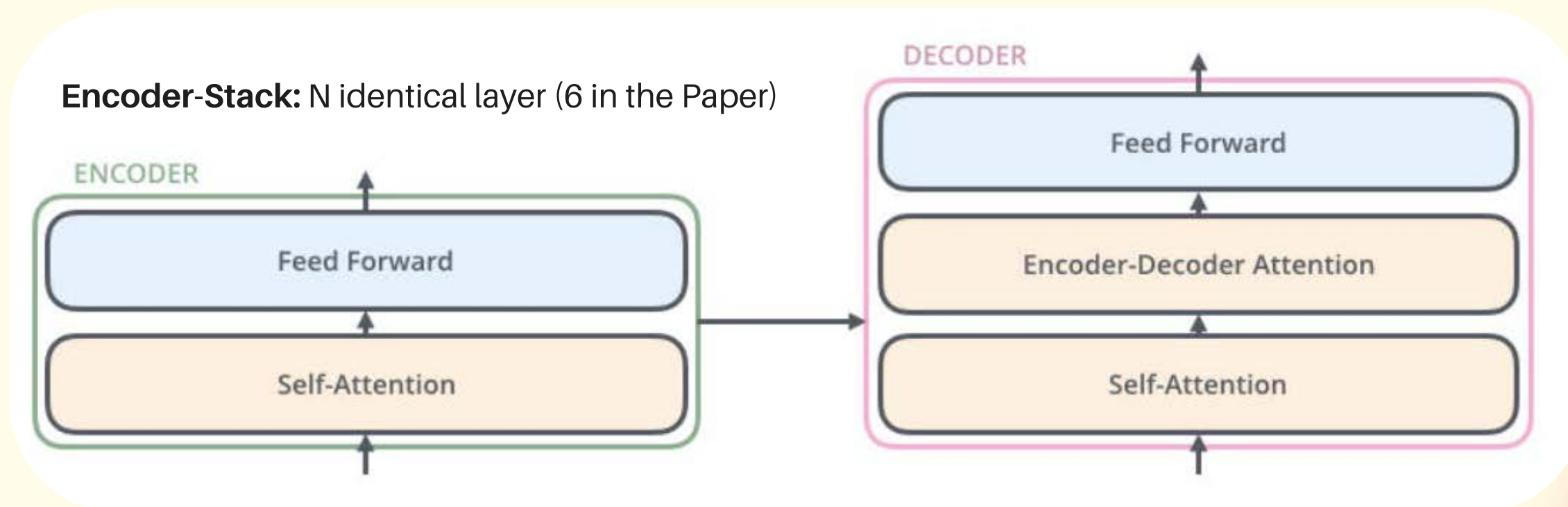
- Replaced the limitations of RNNs and CNNs
- Leverages **attention** mechanisms to capture global dependencies

Architecture

ENCODER-DECODER MODEL

Multi-Head-Attention:

extends self-attention by splitting the input into multiple heads, enabling the model to capture diverse relationships and patterns^[14]



Self-Attention

Lets each part of the input decide **which other parts matter most** for understanding it ^[5]
→e.g. "The dog, who chased the cat, was fast"

To calculate the attention the following abstractions are needed:

- Key (K)
- Value (V)
- Query (Q)

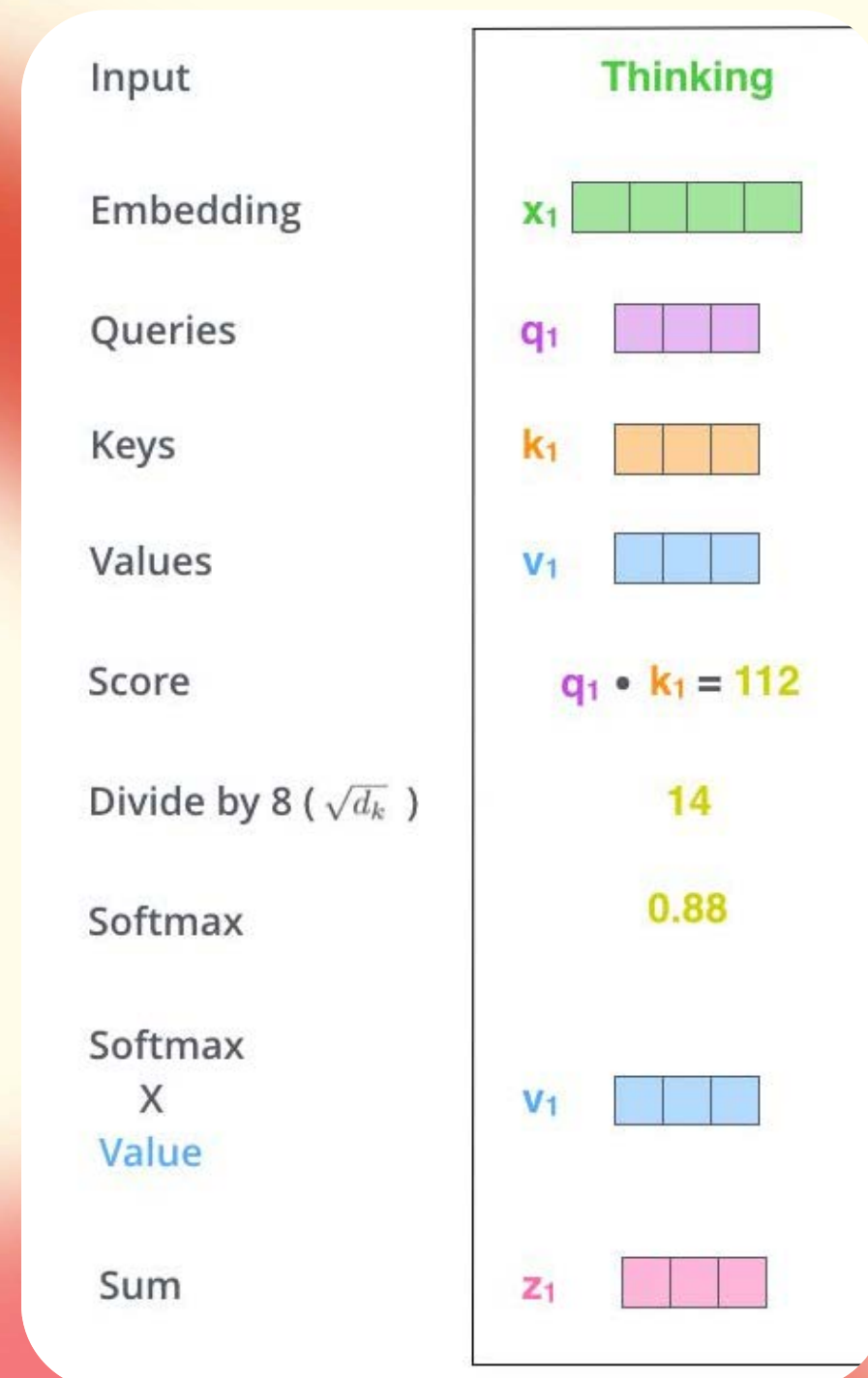
The attention function:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

→ can be described as mapping a query and a set of key-value pairs to an output ^[4]

Self-Attention

IN DETAIL



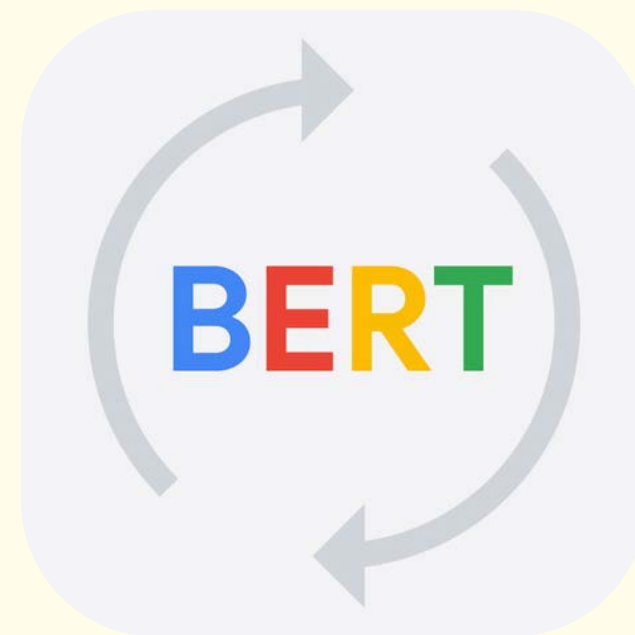
Steps:

1. **K**ey, **V**alue and **Q**uery vector for each input vectors
2. Calculate score = query x key
3. Divide score by key dimension
4. Softmax operation (to normalize the score)
5. Multiply each vector softmax score
6. Sum up weighted value vectors

<https://jalammar.github.io/illustrated-transformer/> ^[5]

Applications

- Natural language processing (**GPT** = Generative Pre-Trained **Transformer**)
- Machine translation
- Text generation
- DNA sequence analysis



APPLYING TO IMAGES

Vision Transformers

“Bridge”

BETWEEN
LANGUAGES AND IMAGES

“An image is worth 16X16 words” (2020 - Google)^[10]



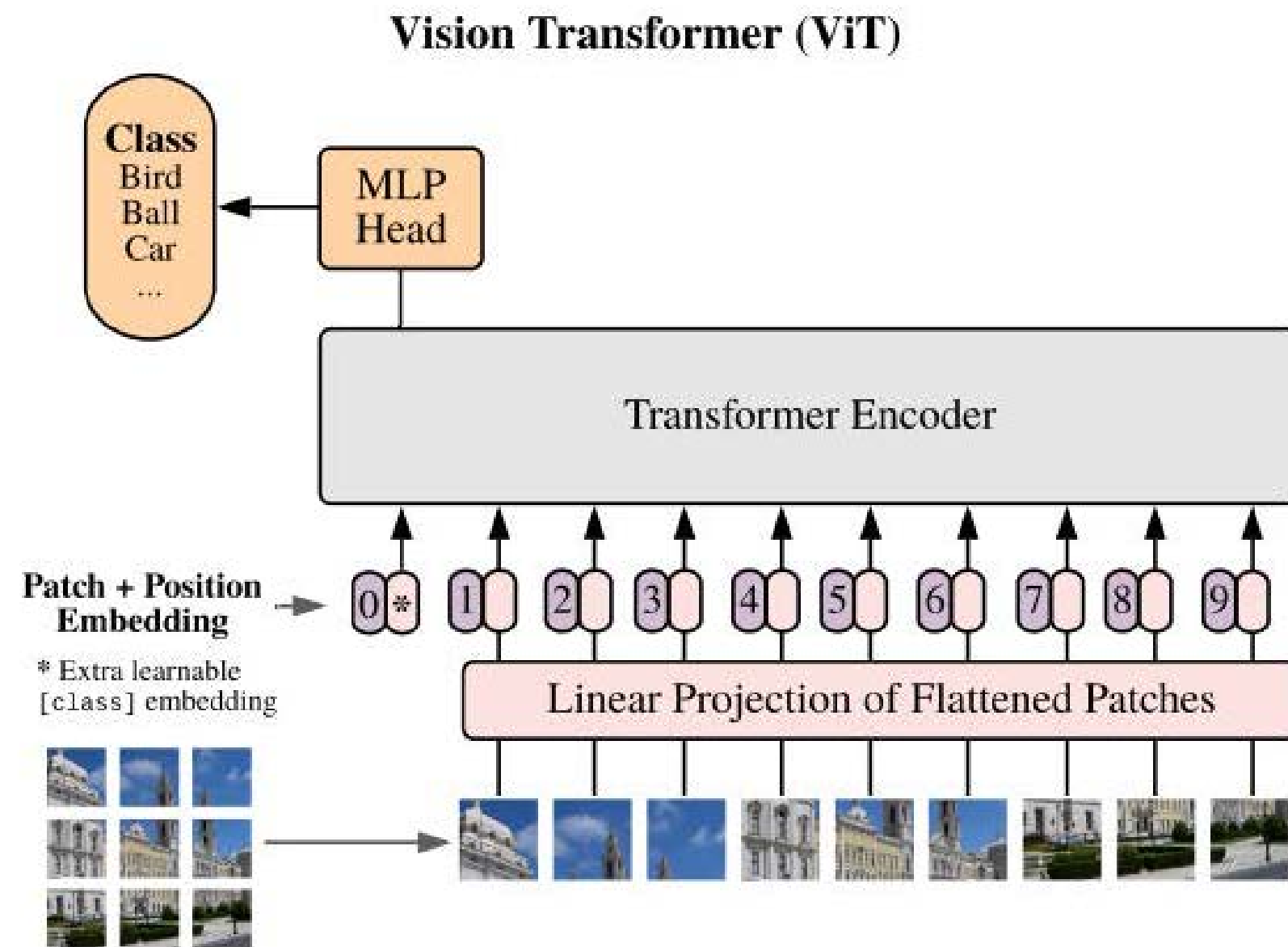
Same architecture works for :

- text (GPT, BERT)
- images (ViT)
- audio
- combinations (multimodal)

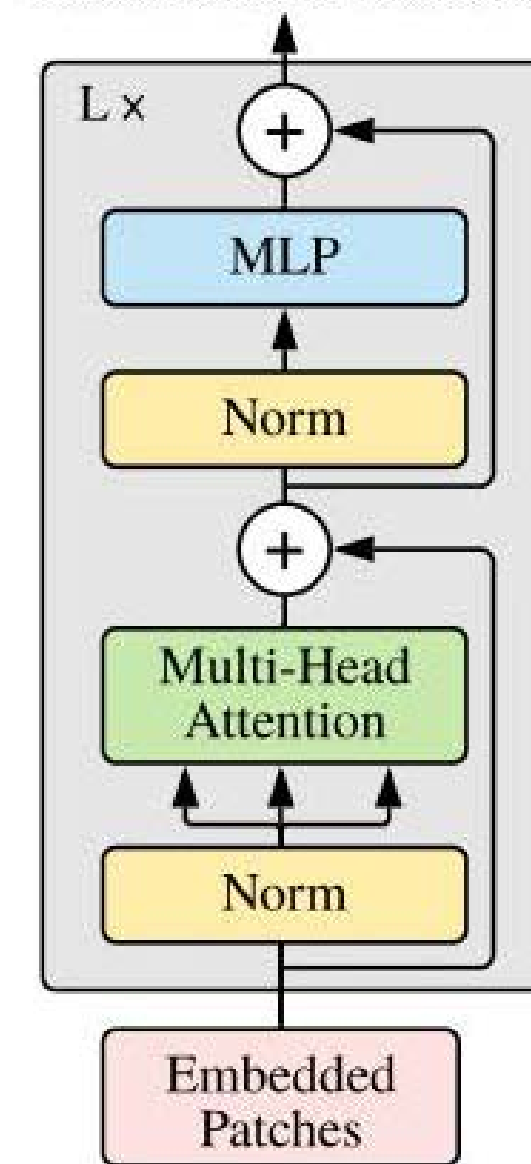
Token Flexibility: both processed with mechanism

- Text has word tokens
- Images have patch tokens

ViT Architecture



Transformer Encoder



Steps:

1. Patchification
2. Embedding
3. CLS Token
4. Transformer Encoder

STEP 1:

Patchification

Sentence to word tokens:

"hi, I am a short sentence"



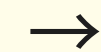
'hi' ',' 'I' 'am' 'a' 'short' 'sentence'

Image to image patches:



splitting an image
into many small squares,
or "patches"

224×224 pixel image



*16×16 pixel patches
= 196 patches total*

STEP 2:

Patch Embedding

a) **Flatten** each patch

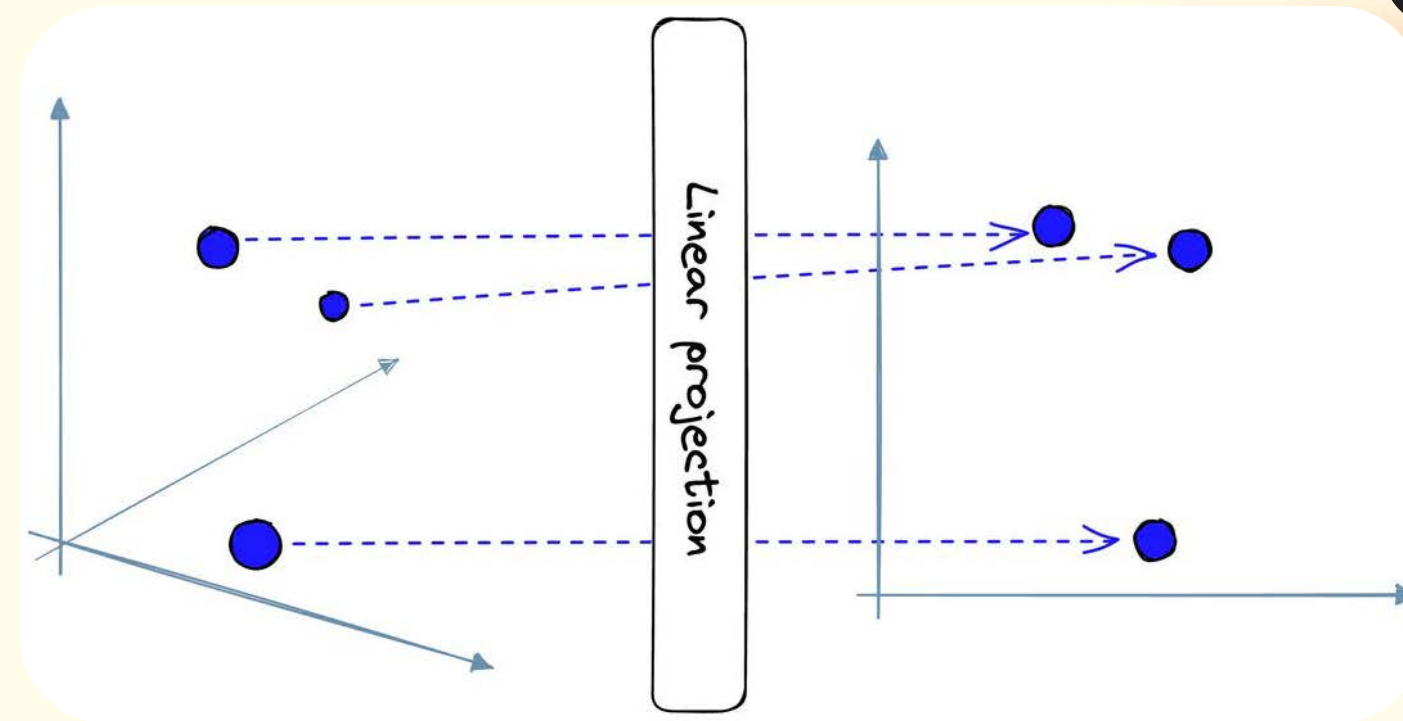
→ turns colored pixels in a patch
into a single long list of numbers

b) **Linearly project** to embedding dimension

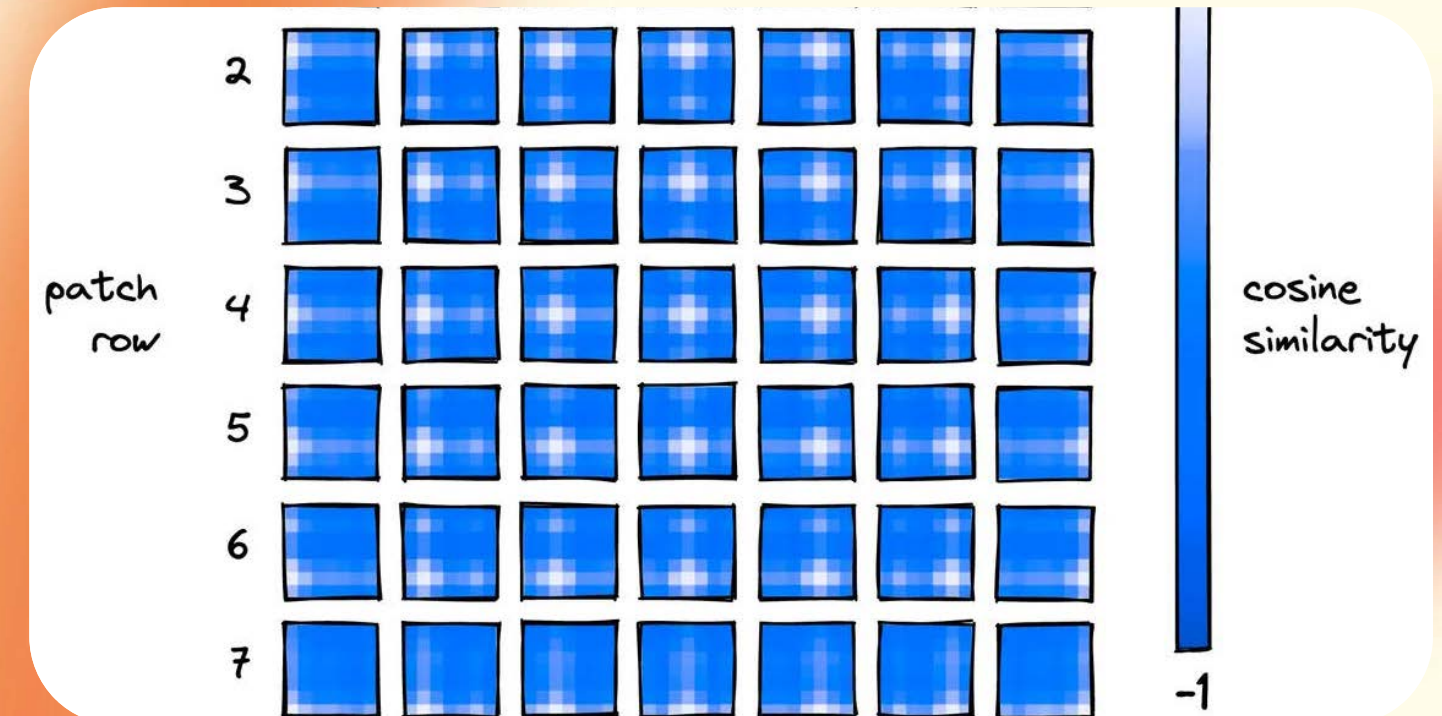
→ maps image patch "arrays" to
patch embedding "vectors"

c) Add **positional encoding** to each patch

→ learned vectors with the same
dimensionality as our patch embeddings



[15]

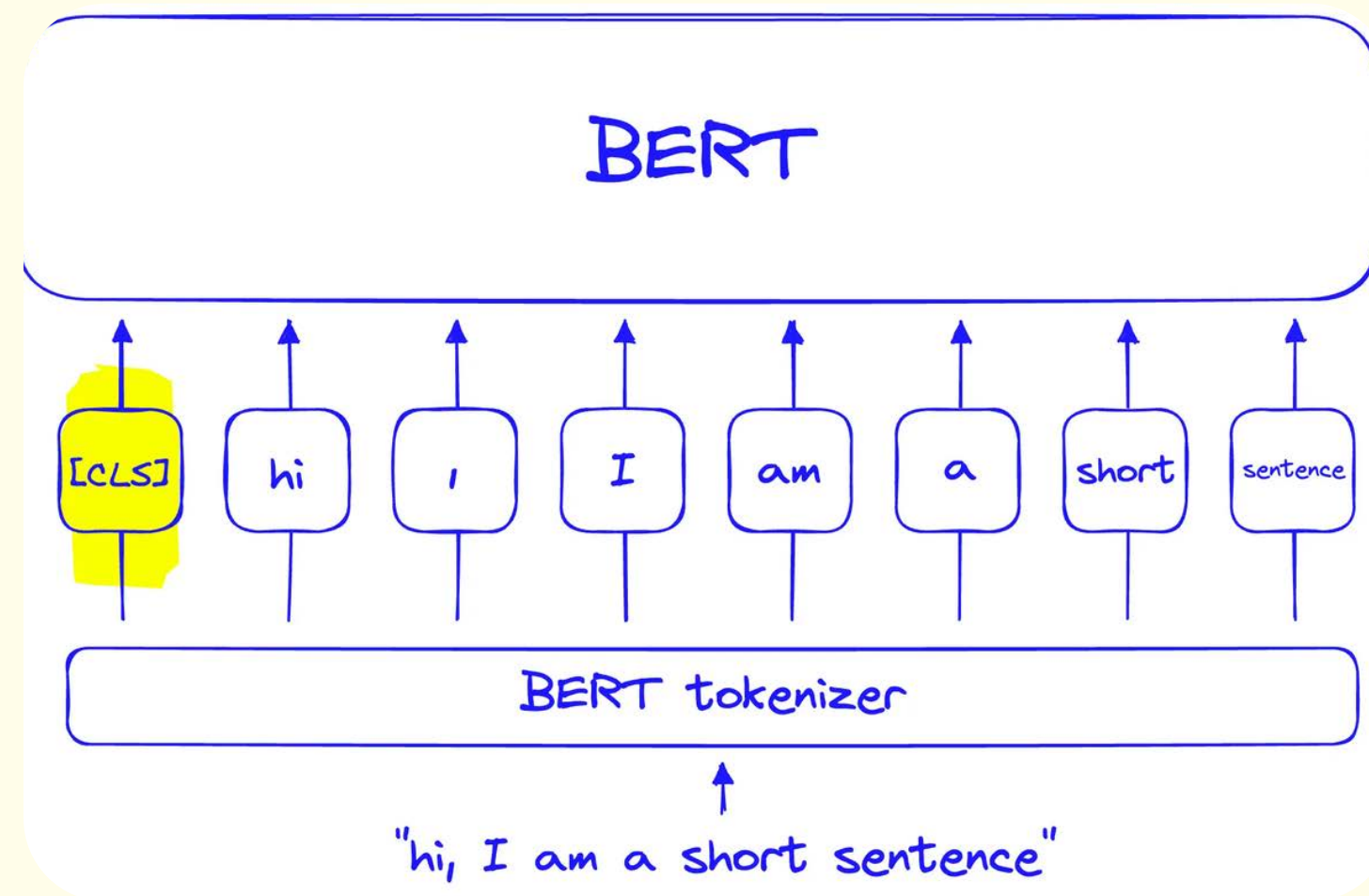


STEP 3:

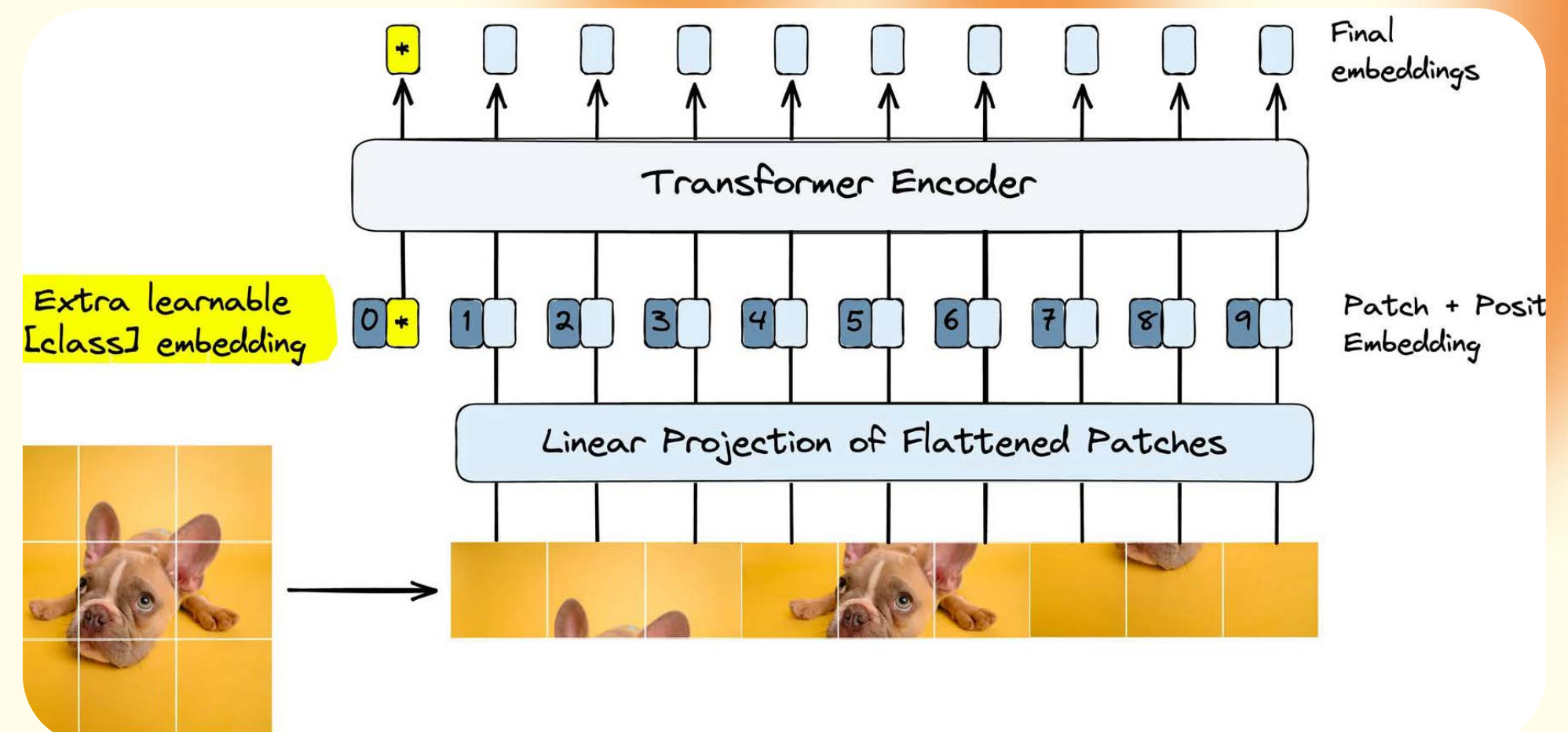
CLS Token

Classification Token = a special, learnable vector

[15]



At the front of the patch sequence



Allows the model to aggregate information from all image patches via self-attention
→ **global representation** of the entire image

Pipeline

Deep Neural Network



input: image



feature-extractor
part

swappable
(e.g. CNN)

+



task-specific

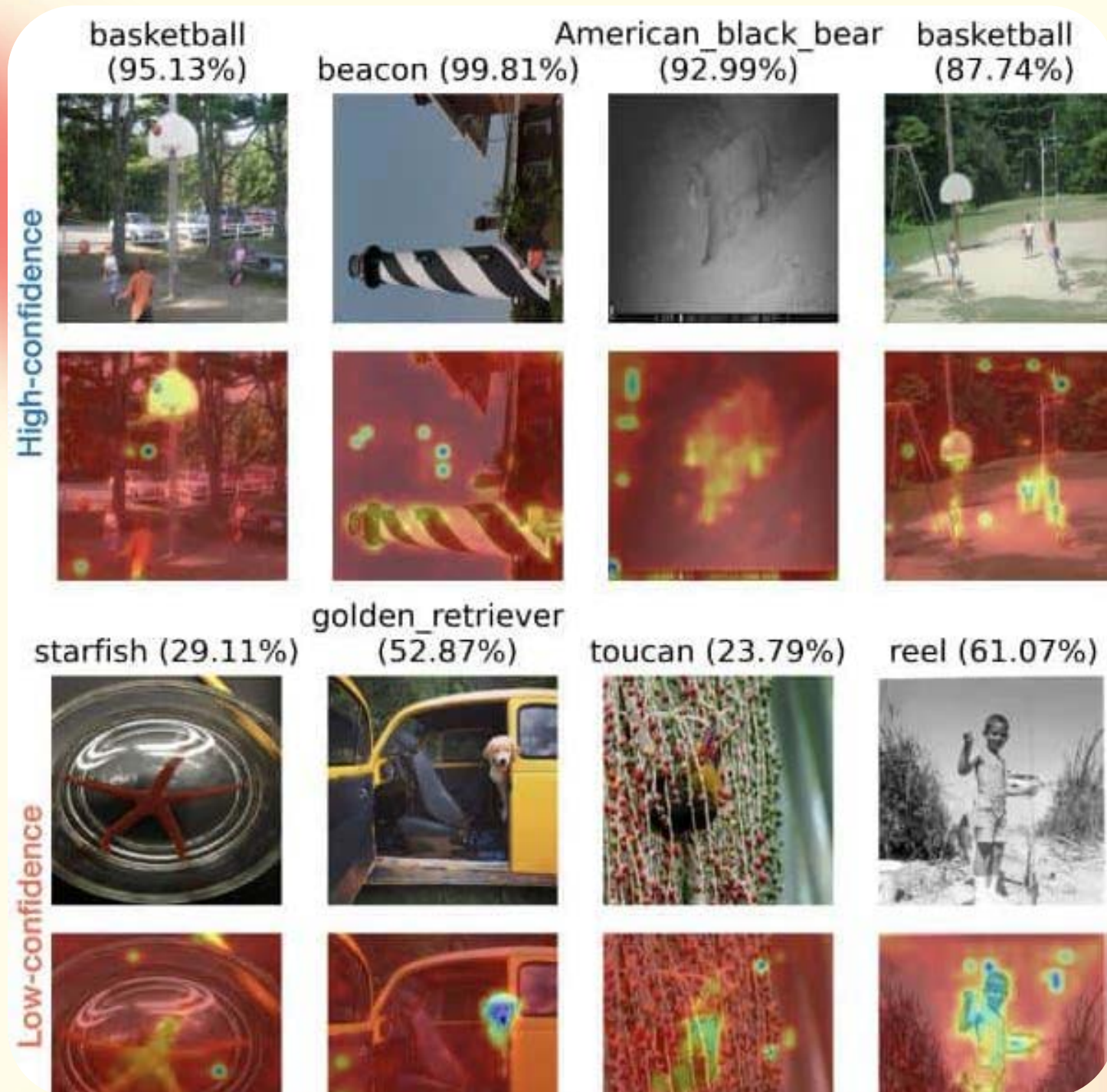
- Class head for classification
- Detection head for detection
- Segmentation decoder for segmentation
- Language decoder for captioning/VQA



output

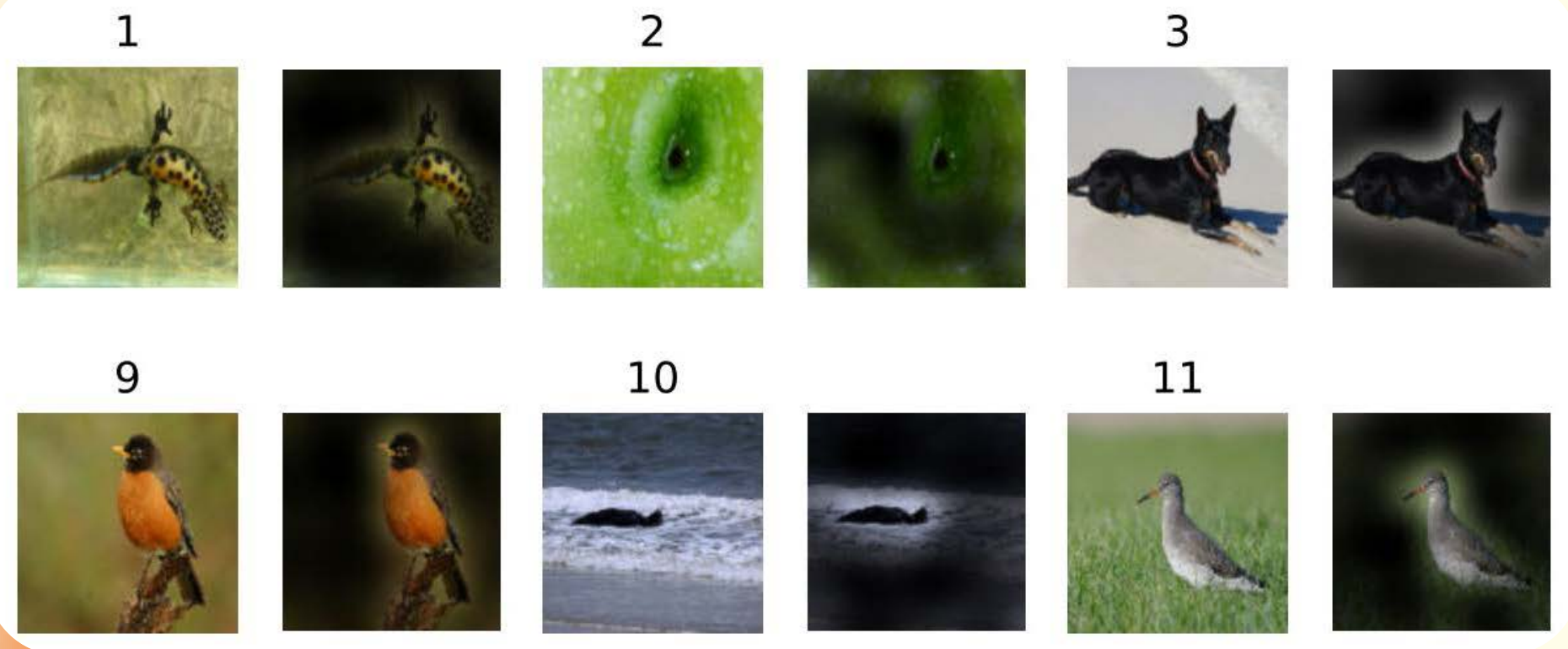
(embeddings/ features)

Attention in ViT



Attention maps =

represent the importance of different parts of an input image to different parts of the model's learned representations^[13]



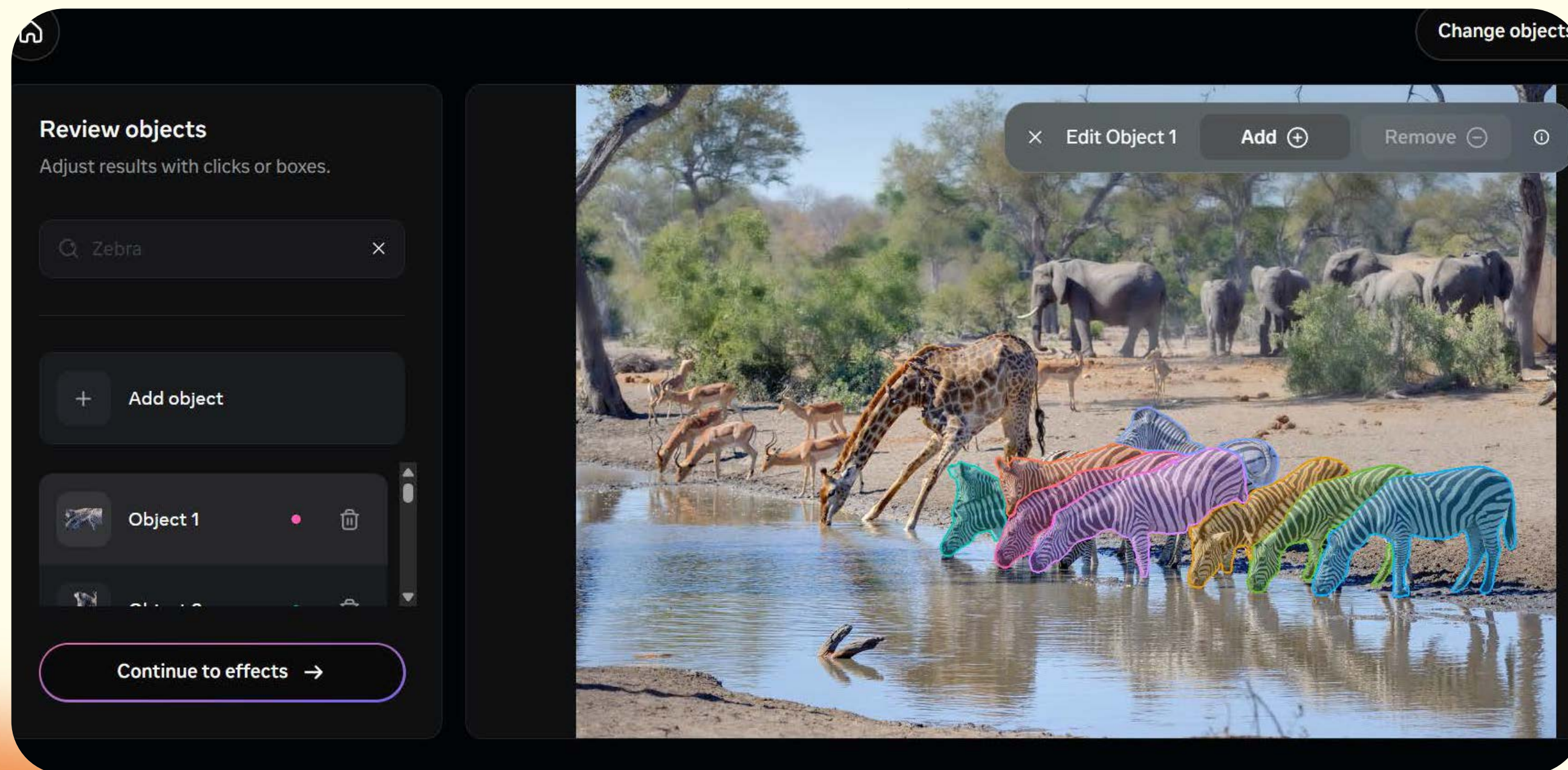
Applications



- Image Classification
- Object Detection
- Image Segmentation
- Generative AI (e.g SORA)
- Autonomous Driving
- Multi-modal Tasks (e.g VQA, CLIP) ^[12,13]



“Segment Anything”



<https://aidemos.meta.com/segment-anything/gallery/>

Practical Task

"SEGMENT ANYTHING"

1. Choose a variety of different images (*e.g. street scene, group of people, nature scene*)
2. Try out the following:
 - a. Single-object segmentation
 - b. Segment "everything"
 - c. Text-based segmentation
3. Reflect:
 - a. What worked the best and what difficulties occurred?
 - b. Where do you see potential real-world applications?

Limitations

- **Data Hungry**
 - perform best with large datasets (1M+ images)+
 - Poor on small datasets: without pre-training, performs worse than CNNs
- **Expensive** computing costs ^[16]
- **Black box**: hard to understand and interpret
- **Fine-grained details**: struggles on high-resolution

Comparison

Aspect	CNN	ViT
Processing	Local convolution filters	Patches + global attention
Data efficiency	Works with <10k images	Needs 100k+ or pretraining
Speed	Fast training & inference	Slower initially, parallelizable
Global context	Hierarchical, builds gradually	Immediate via self-attention
Compute	Low	High ($O(n^2)$)
Transfer learning	Good	Excellent with pretraining
Interpretability	Filters are visual	Attention is abstract

Conclusion

→ Depending on the use case, resources and task

CNN:

- Efficient, stable with small datasets, mature tooling
 - Detection/segmentation via feature pyramids (e.g., FPN)
- Limited data/compute, latency-critical deployments

ViT:

- Captures long-range dependencies, strong transfer with big pretraining, fits multimodal setups
 - Classification via CLS, detection/segmentation with decoders
- Abundant pretraining, tasks needing global context, multimodal (vision+language) work

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ChatGPT with tutoring prompt used for structuring the presentation and support to understand the matter

**Thank you
for your attention!**

ANY QUESTIONS?